Adopted Levels

Type	Author	History Citation	Literature Cutoff Date
Full Evaluation	Balraj Singh	NDS 156, 148 (2019)	31-Jan-2019

Q(β^-)=−5330 *SY*; S(n)=7220 *SY*; S(p)=3040 *SY*; Q(α)=10774 *15* 2017Wa10
Estimated uncertainties (2017Wa10): 1050 for Q(β^-), 1010 for S(n), 1130 for S(p).
Q(α)=10.776 MeV *15* from Eα=10.628 *15* (2016Ho09, 2015Og05, see also other review articles from the Dubna and GSI groups).
S(2n)=13100 *1040* (syst,2017Wa10). S(2p)=4120 (theory,1997Mo25).
EVR=Evaporation Residues.

The ^{292}Lv nuclide has been identified by Dubna-Livermore-ORNL collaboration (2004Og12, also many other papers from this group), and later independently by the GSI group (2012Ho12). A total of ten EVR-α-α-SF correlated decay chains have been reported so far. The ^{292}Lv nuclide decays 100% (2016Ho09) by α to ^{288}Fl. 2017Ka66 (also 2014MoZV) reported observation of three correlated events at GARIS-RIKEN facility, one with a tentative identification of EVR-α-α-α-SF event starting from ^{292}Lv and ending in ^{280}Ds, which decays by SF. No information is yet available from x-ray observation or chemical procedures.

Experiments at FLNR-JINR-Dubna, using Dubna gas-filled recoil separator (DGFRS):
2004Og12, 2004OgZZ (also 2005Og03,2007Og05,2009Og07,2011Og07): ^{292}Lv produced and identified in ^{248}Cm(^{48}Ca,4n), E=247 MeV, excitation energy of 36.8-41.1 MeV for the compound nucleus ^{296}Lv. Experiments carried out using U400 cyclotron and Dubna gas-filled recoil separator (DGFRS) at FLNR-JINR-Dubna facility. Identification of mass number and nuclide was made on the basis of excitation functions, and detecting correlated events between evaporation residues (EVR), alpha particles and SF decays. A total of six decay chains were assigned to ^{292}Lv, but in one of these events EVR was missing, and in two events, one of the alphas escaped detection, as summarized below. Measured cross sections, EVR-α-α-SF correlations, Eα, half-lives. Note that an earlier assignment of one EVR-multiple α-SF correlated decay chain assigned to the ^{292}Lv nuclide was by the Dubna-Livermore collaboration in 2001Og01 and 2002Og09 (also 1999Og10,2000Og07, 2001Og06,2001Og11,2002Og03,2002Og09,2002Og13) by interpreting 4n-channel in ^{248}Cm(^{48}Ca,xn)^{296}Lv*,E(^{48}Ca)=240 MeV reaction, and compound nucleus excitation energy of 30.4-35.8 MeV. Later this decay chain was assigned by 2004Og07 and 2004Og12 (also 2005Og03) to 3n-reaction channel, instead, thus leading to the discovery of the ^{293}Lv nuclide (see ^{293}Lv Adopted Levels).
Additional information 1.

History of EVR -> α1 -> α2 -> SF decay chains from ^{248}Cm(^{48}Ca,4n)^{296}Lv*, E=247 MeV (2004OgZZ), where Δt values are time differences between successive decays:

Chain #1: Energy of EVR=10.6 MeV:
$E_{\alpha 1}$=10.63 MeV *8*, Δt_1=20.7 ms, assigned to ^{292}Lv.
$E_{\alpha 2}$=9.99 MeV *8*, Δt_2=2.239 s, assigned to ^{288}Fl.
E_{SF}=178 MeV, Δt_3=24.599 ms, assigned to ^{284}Cn: decays by SF.

Chain #2: Missing EVR.
$E_{\alpha 1}$=10.63 MeV *14*, assigned to ^{292}Lv.
$E_{\alpha 2}$=10.01 MeV *17*, Δt_2=0.832 s, assigned to ^{288}Fl.
E_{SF}=201 MeV, Δt_3=54.8 ms, assigned to ^{284}Cn: decays by SF.

Chain #3: Energy of EVR=12.6 MeV:
$E_{\alpha 1}$=10.68 MeV *7*, Δt_1=35.2 ms, assigned to ^{292}Lv.
$E_{\alpha 2}$=missing α, assigned to ^{288}Fl.
E_{SF}=193 MeV, Δt_3=616.7 ms, assigned to ^{284}Cn: decays by SF.

Chain #4: Energy of EVR=12.1 MeV:
$E_{\alpha 1}$=10.66 MeV *12*, Δt_1=11.832 ms, assigned to ^{292}Lv.
$E_{\alpha 2}$=9.82 MeV *12*, Δt_2=1.090 s, assigned to ^{288}Fl.
E_{SF}=213 MeV, Δt_3=274.5 ms, assigned to ^{284}Cn: decays by SF.

Adopted Levels (continued)

Chain #5: Energy of EVR=14.6 MeV.
$E_{\alpha 1}$=missing α, assigned to ^{292}Lv.
$E_{\alpha 2}$=9.85 MeV *10*, Δt_2=0.416 s, assigned to ^{288}Fl.
E_{SF}=212 MeV, Δt_3=158.1 ms, assigned to ^{284}Cn: decays by SF.

Chain #6: Energy of EVR=14.4 MeV:
$E_{\alpha 1}$=10.69 MeV *47*, Δt_1=35.9 ms, assigned to ^{292}Lv.
$E_{\alpha 2}$=9.91 MeV *14*, Δt_2=0.509 s, assigned to ^{288}Fl.
E_{SF}=202 MeV, Δt_3=117.9 ms, assigned to ^{284}Cn: decays by SF.

Experiments at GSI using SHIP recoil separator:

2012Ho12: ^{248}Cm(^{48}Ca,4n)^{296}Lv*, E=265.4,270.2 MeV, 40.9 MeV excitation energy of the compound nucleus, SHIP-GSI facility. Measured cross sections, (EVR)-α-α-SF position and time correlations, Eα, half-lives, Eγ. Four correlated decay chains were observed. See also 2016Ho06 for analysis of shell-correction energies and estimates of fission barriers using experimental data for several decay chains, including that of ^{292}Lv.

History of EVR -> α1 -> α2 -> SF decay chains from ^{248}Cm(^{48}Ca,4n)^{296}Lv*, E=265.4 MeV (2012Ho12):

Chain #1: Energy of EVR=9.0 MeV:
$E_{\alpha 1}$=2.715 MeV, Δt_1=3.3 ms, assigned to ^{292}Lv.
$E_{\alpha 2}$=9.959 MeV *15*, Δt_2=994 ms, assigned to ^{288}Fl.
E_{SF}=207 MeV *20*, Δt_3=50 ms, assigned to ^{284}Cn: SF decay.
Eγ=180.3 keV seen in coincidence mode.

Chain #2: Energy of EVR=18.0 MeV:
$E_{\alpha 1}$=10.625 MeV *15*, Δt_1=0.076 ms, assigned to ^{292}Lv.
$E_{\alpha 2}$=9.896 MeV *15*, Δt_2=1.236 ms, assigned to ^{288}Fl.
E_{SF}=220 MeV *20*, Δt_3=269 ms, assigned to ^{284}Cn: SF decay.
Eγ=55.0 keV seen in coincidence mode.

Chain #3: Energy of EVR=22.3 MeV:
$E_{\alpha 1}$=1.4 MeV, Δt_1=28.8 ms, assigned to ^{292}Lv.
$E_{\alpha 2}$=1.8 MeV, Δt_2=252 ms, assigned to ^{288}Fl.
E_{SF}=218 MeV *20*, Δt_3=121 ms, assigned to ^{284}Cn: SF decay.
Eγ=350.4 keV seen in coincidence mode.

Chain #4: Energy of EVR=20.8 MeV:
$E_{\alpha 1}$=10.630 MeV *50*, Δt_1=11.6 ms, assigned to ^{292}Lv.
$E_{\alpha 2}$=9.837 MeV *50*, Δt_2=72 ms, assigned to ^{288}Fl.
E_{SF}=206 MeV *20*, Δt_3=25 ms, assigned to ^{284}Cn: SF decay.
Eγ=496.1 keV seen in coincidence mode.

2017Ka66: ^{292}Lv produced in ^{248}Cm(^{48}Ca,4n),E=261.6 MeV (E^*=41.3 MeV using RIKEN Linear Accelerator (RILAC) and gas-filled recoil ion separator (GARIS). Two EVR-α-α-SF correlated events and a third tentative EVR-α-α-α-SF were observed with the decay history given below. Production σ=3.1 pb *+28−18* for three chains. The third chain, with two possible interpretations, also led to a tentative identification of ^{280}Ds isotope. 2014MoZV is an earlier report from the same group.
Chain #1: Energy of EVR=15.6 MeV.
$E_{\alpha 1}$=10.79 MeV *7*, Δt_1=32 ms, assigned to ^{292}Lv.
$E_{\alpha 2}$=9.89 MeV *3*, Δt_2=548 ms, assigned to ^{288}Fl.
E_{SF}=232 MeV, Δt_3=65 ms, assigned to ^{284}Cn: SF decay; SF decay events in coincidence with 371.7-keV γ ray.

Chain #2: Energy of EVR=12.9 MeV.
$E_{\alpha 1}$=2.77 MeV escaped α, Δt_1=2.3 ms, assigned to ^{292}Lv.

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$E_{\alpha 2}$=9.99 MeV *3*, Δt_2=243 ms, assigned to ^{288}Fl.
E_{SF}=182 MeV, Δt_3=839 ms, assigned to ^{284}Cn: SF decay.

Chain #3 (first interpretation): Energy of EVR=15.3 MeV:
$E_{\alpha 1}$=10.66 MeV *2*, Δt_1=13.1 ms, assigned to ^{292}Lv.
$E_{\alpha 2}$=3.50 MeV escaped α, Δt_2=282 ms, assigned to ^{288}Fl.
E_{SF}=163 MeV, Δt_3=9.6 ms, assigned to ^{284}Cn: SF decay; SF decay events in coincidence with 393.1-keV γ ray.

Chain #3 (second interpretation): Energy of EVR=15.3 MeV:
$E_{\alpha 1}$=10.66 MeV *2*, Δt_1=4.1 ms, assigned to ^{292}Lv.
$E_{\alpha 2}$=0.83 MeV escaped α, Δt_2=9.0 ms, assigned to ^{288}Fl.
$E_{\alpha 3}$=3.50 MeV escaped α, Δt_3=282 ms, assigned to ^{284}Cn.
E_{SF}=163 MeV, Δt_4=9.6 ms, assigned to ^{280}Ds: SF decay; SF decay events in coincidence with 393.1-keV γ ray.

For theoretical studies, consult Nuclear Science References (NSR) database at NNDC, BNL for 233 primary references dealing with the half-lives and other aspects of nuclear structure in this mass region.

^{292}Lv Levels

E(level)	J^π	$T_{1/2}$	Comments
0	0^+	12.8 ms *+70−33*	%α=100 (2016Ho09); %SF≤9 (2016Ho09) Only the α decay mode has been observed. E(level): the reported activity is assumed to belong to the g.s. of ^{292}Lv. $T_{1/2}$: from analysis of eight decay chains (2012Ho12, also 2016Ho09, 2015Og01, 2015Og05, 2015OgZX, and 2017Og01 review articles), value is based on the analysis of a total of ten observed events (six by Dubna-Livermore collaboration and four by the GSI group) out of which four from each experiment were used for the determination of average half-life. Value of 11.9 ms *+217−26* from two correlated events in 2017Ka66 is not included in this averaged value, however, its inclusion will hardly affect the adopted value. Other: $T_{1/2}$=18 ms *+16−6* (2004Og12,2011Og07) from four out of six observed events assigned to ^{292}Lv. Theoretical Q(α)=10.82 MeV, and α-decay half-life=43 ms (1997Mo25). Only the α-decay mode has been observed. Out of a total of ten events assigned to ^{292}Lv, alpha-particle energies were analyzed from seven events, with an average Eα=10.628 MeV *15* (2016Ho09,2015Og01) deduced from four events, implying Q(α)=10.776 MeV *15* for the decay of ^{292}Lv populating the g.s. in ^{288}Fl. Other Eα=10.66 MeV *7* (2004Og12,2011Og07) from five out of six observed events assigned to ^{292}Lv, 10.625 MeV *15* (2012Ho12), 10.67 MeV *2* (2017Ka66, from two correlated events). Measured production σ=3.3 pb *+25−14* at E(^{48}Ca)=247 MeV (2004Og12); 3.4 pb *+27−16* at E(^{48}Ca)=265.4 MeV (2012Ho12); 3.1 pb *+28−18* at E(^{48}Ca)=216.6 MeV (2017Ka66).